Hiking speed from GPS traces and remote sensing

Status since 9 Oct: T1 complete, T2 tier 1, first T3 results

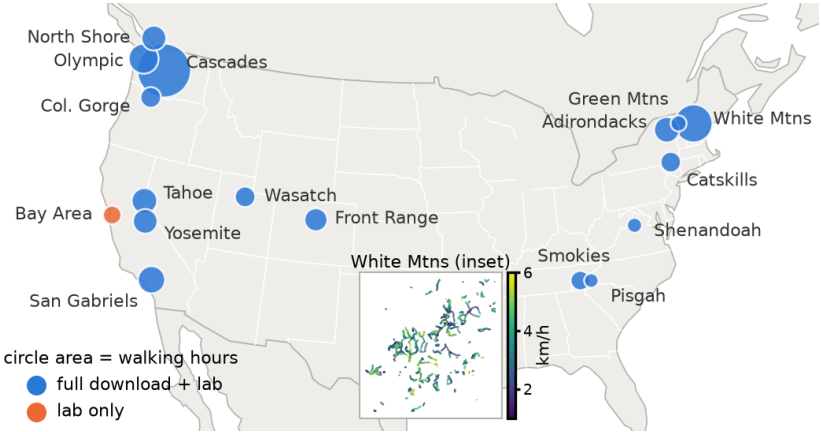
Brian Zhou
mentors Danni Beaulieu, Tianyi Gao; advisor Prof. Nathan Jacobs

16 Oct 2026

Since 9 Oct

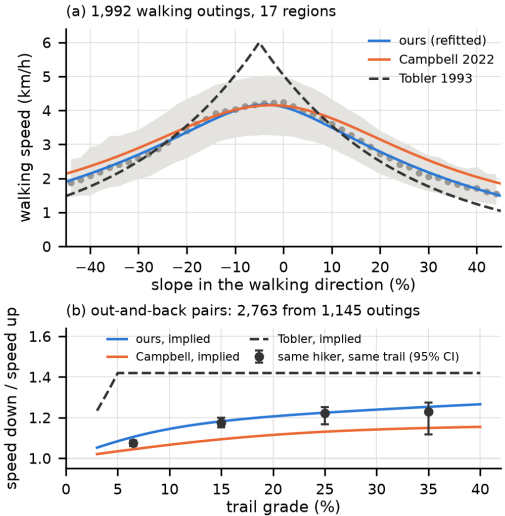
- **T1 complete: 17 regions** (US and Canada). 16 from full OSM downloads plus the lab crawl; the Bay Area from the crawl only.
  - 2,876 hiking traces; walking set: 1,992 outings, 4,702 moving hours
  - 290,640 walking segments of 50 m, one combined dataset and data card
- **T2 tier 1 done for all regions**: terrain from 5 m and 1 m elevation, 30 m vegetation, AlphaEarth satellite embeddings.
- **T3 evaluation harness**: every model predicts a speed *range* per segment and is graded only on hikes, map squares or **whole regions** it never saw (CRPS, coverage, hike time).
- **Lead result**: terrain from remote sensing transfers to unseen regions; map tags barely do.

The dataset



Circle area = walking hours. Inset: White Mountains segments by speed.

Speed vs slope: Tobler is too fast and too lopsided



- Measured speed peaks near **4.2 km/h** on level to gently descending trail (Tobler: 6.0).
- Tobler: ∼20% too fast on gentle grades, ∼18% too slow on climbs over 25%.
- Campbell 2022: right near level, ∼20% too fast on steep climbs.
- Same hiker, same trail (2,763 out-and-back pairs): downhill is 8% faster than uphill at 3–10% grade, 22% above 20%. Tobler says 42% everywhere.

How we test

Three ways to hide data

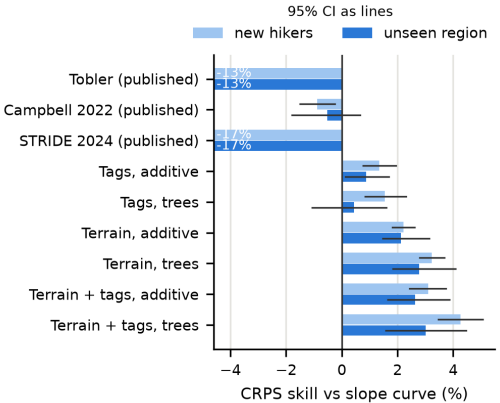
- new hikers (5 folds of outings)
- new places (5 folds of 10 km squares)
- **a new region** (leave one of 17 out)

Score: CRPS skill vs a slope curve refitted to our data; 95% intervals resample whole regions.

Models

- published: Tobler 1993, Campbell et al. 2022, STRIDE 2024
- slope curve (baseline)
- + OSM tags / + terrain / + embeddings / + vegetation (additive model and boosted trees)
- references: the hiker's own pace (first km), the region's own offset

Lead result: terrain transfers, tags don't

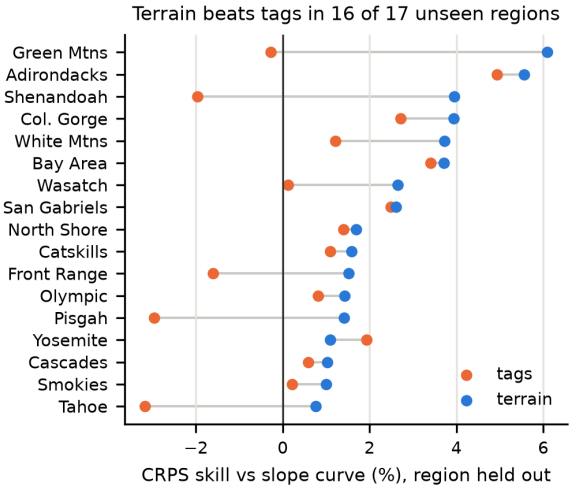


Unseen region	skill [95% CI]
Tobler 1993	−12.5%
STRIDE 2024	−16.8%
Campbell 2022	−0.5%
OSM tags	+0.9% [0.1, 1.7]
Embeddings	+1.1% [0.1, 2.6]
Terrain	+**2.1**% [1.5, 3.1]
Terrain (trees)	+**2.8**% [1.8, 4.1]
Terrain + emb. + tags (trees)	+3.9% [2.7, 5.7]

Terrain beats tags in **16 of 17** held-out regions.

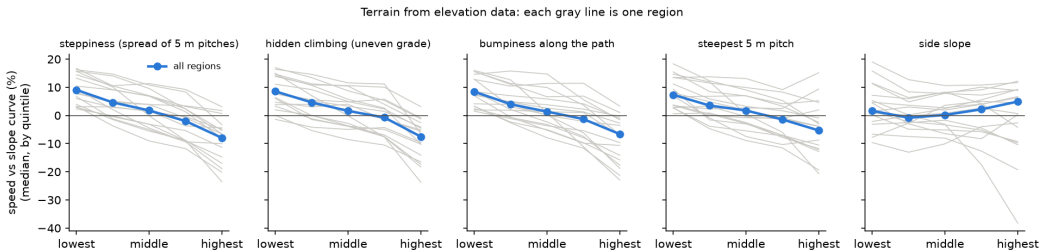
All models get their range the same way: prediction + error spread on held-out training hikes.

Per region: terrain beats tags in 16 of 17



- Terrain minus tags: +0.7 to +2.2 points (95%), above zero in every bootstrap draw.
- In 5 regions tags make predictions *worse* (Tahoe, Pisgah, Shenandoah, Front Range, Green Mtns); terrain helps in all 17.
- The exception: Yosemite (tags +1.9, terrain +1.1).
- Why tags fail: within a hike they act the same everywhere; but sac_scale is on 1–96% of trail by region, and "no tag" means different things in different regions.

What “terrain” means

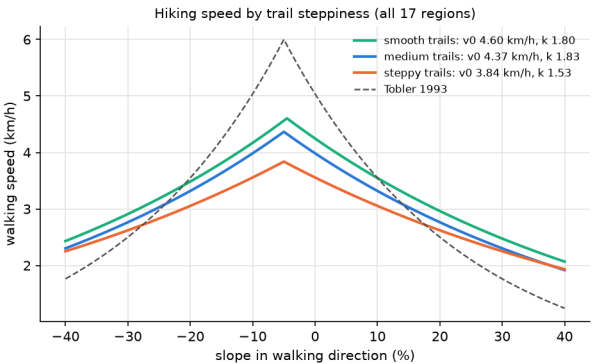


Read every 1 m along the matched path from the elevation grid. Gray: one line per region; blue: all regions. Steppier trail profiles are slower everywhere: smoothest fifth +9%, steppiest fifth −8% vs slope alone. The gain in unseen regions is largest on descents and level trail (+2.3 to +2.8%), smallest on steep climbs (+0.8%): footing limits descents, effort limits climbs.

Checks

- **Not a region proxy**: terrain still adds $\sim$1.9 points over each region's own speed offset.
- **Not a hiker proxy**: still $\sim$2.3 points once the hiker's pace is known.
- **No neighbours in training**: a whole zone held out, terrain $+1.9\%$ (tags $+0.7\%$); West $\leftrightarrow$ East $+2.6\%$ (tags $+1.5\%$). **Newer hikes** $+2.2\%$. **Season, time of day, weather** add nothing in new regions.
- **GPS rate**: the same gain for traces recorded every $\leq$2 s, 2–10 s, $>$10 s ($+1.7$, $+2.4$, $+1.9\%$). Sparse traces read $\sim$10% slow; thinning dense walks reproduces it (cut corners, hidden stops).
- **Hidden climbing**: $\sim$60% of the gain is uneven grade inside the 50 m segment; the rest is difficulty beyond grade.
- **1 m lidar $\approx$ 5 m** for this signal; STRIDE's roughness adds nothing on trails.
- **30 m vegetation adds nothing**; **embeddings** help within regions ($+2.6\%$) but transfer less reliably ($+1.1\%$).
- **Calibration**: 90% ranges hold 88–90%; a terrain-dependent spread doesn't help.
- **Mode labels**: 5% of "walkers" are labeled bike/ski by their own app (327 labeled traces).

A terrain-adjusted Tobler formula

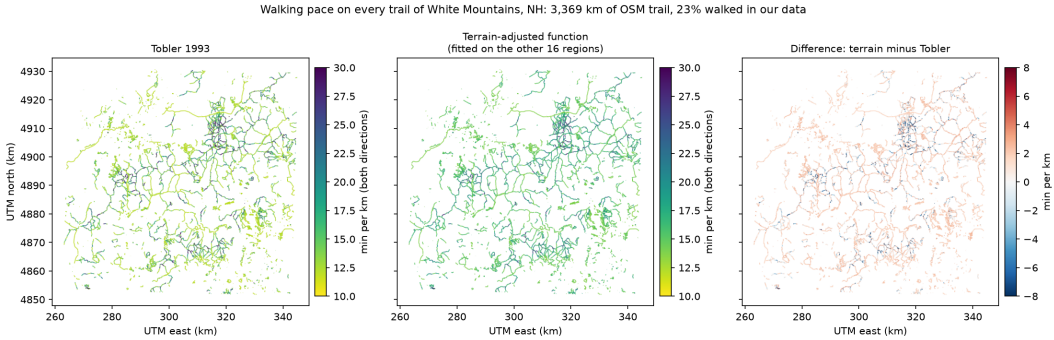


$$\ln v = a_0 + a_1 z - (k_0 + k_1 z)\,|s + s_0|$$

z = spread of 5 m pitches / 0.10 (one number from any DEM).

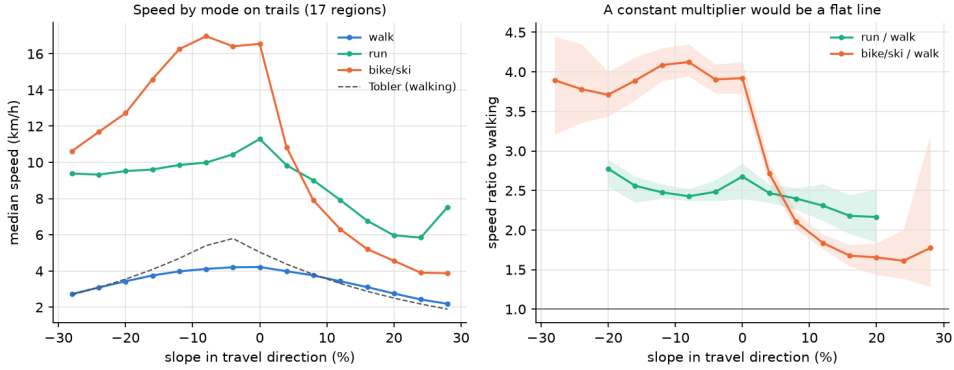
- Peak: smooth trails 4.60, steppy 3.84 km/h (Tobler: 6.0).
- +1.0% [0.6, 1.5] in unseen regions with 5 numbers; less portable than the full models (+0.6% for an unseen zone).
- A drop-in for GIS tools that use Tobler today.

Pace on every White Mountains trail



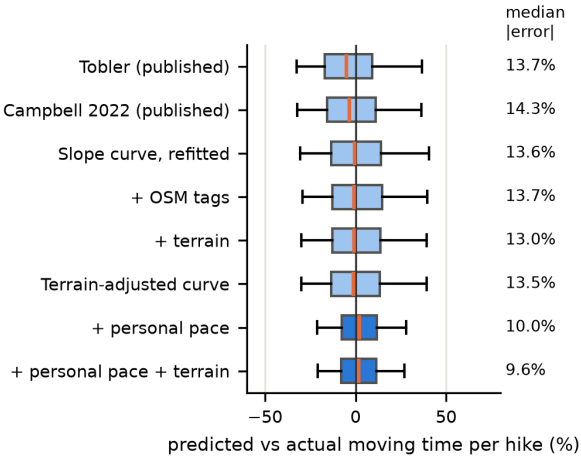
3,369 km of OSM trail, 77% never walked in our data. The formula is fitted on the other 16 regions. Right: difference from Tobler.

Travel modes on the same trails



Bike/ski $\approx 4\times$ walking on descents but $\approx 1.6\times$ on climbs, so a constant mode multiplier fails. Runners $\approx 2.5\times$. On held-out hikes, a constant $\times 2.5$ is as good as a runner curve of its own; bikes and skis need their own (3.8% better).

Whole hikes: the hiker matters most



Moving time per hike, 1,354 hikes in held-out regions (median error):

- slope curve 13.6%
- + terrain 13.0%
- + the hiker's own pace (first km) 10.0%
- + own pace + terrain **9.6%**; about half of hikes within ±10%

Questions

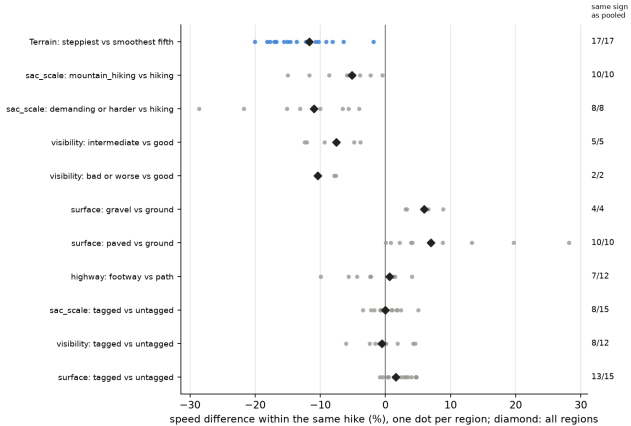
1. **Paper framing**: lead with "terrain from remote sensing transfers to new regions; map tags don't", with Tobler, Campbell and STRIDE as baselines. Agree?
2. **Embeddings**: present AlphaEarth as "helps locally, transfers less"? Try another embedding?
3. **Lidar understory** (STRIDE's 0.85–1.2 m band): $\sim$12 GB for the White Mountains alone. 1–3 regions, or skip?
4. **Data release**: a de-identified copy is ready; license (ODbL) and lab policy?
5. **Cluster account**: `engr-class-any` or the lab's?
6. **Sparse GPS traces** read up to 11% slow (how we measure). Fix in a T1 v2, use only traces logged every $\leq$10 s for absolute speeds, or report it?
7. **Figures**: the draft is at 5 pages with 6 figures. Which 3–4 to keep?

Next: paper draft (skeleton with table and figures exists; full draft by $\sim$27 Nov; deadline 11 Jan).

Backup slides

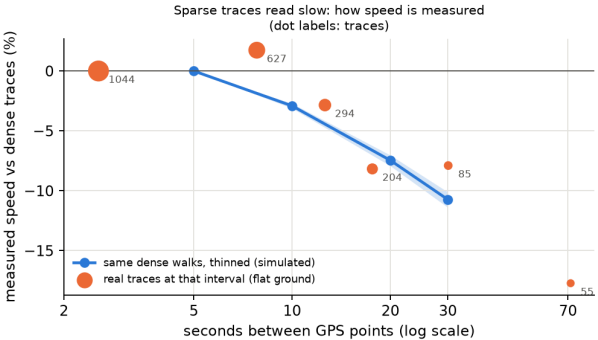
only if asked

Backup: why tags fail



Within the same hike (same hiker, same day), terrain and tag values act the same in every region. Whether a tag is there at all does not: sac_scale is on 1% (Green Mtns) to 96% (Adirondacks) of walked trail.

Backup: sparse GPS traces read slow



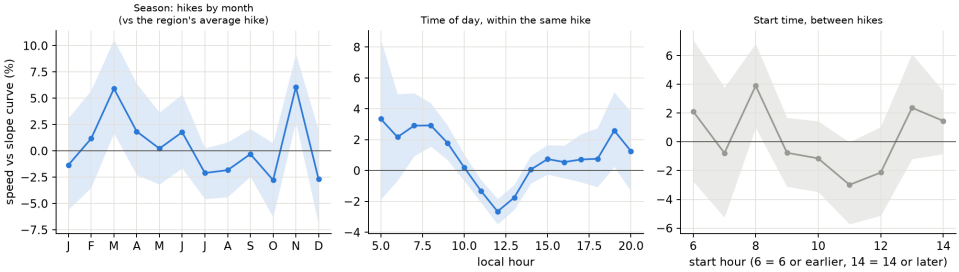
- 1,020 dense walks thinned to 30 s read 11% slow: -6% distance (straight lines cut corners), $+8\%$ moving time (short stops hide).
- Real sparse traces show the same.
- Terrain's gain is the same at every rate; the curve without >10 s traces peaks 3% higher.

Backup: no neighbouring region in training

Skill vs slope curve (%)	Leave one region out	Whole zone out	West ↔ East
Terrain, simple / trees	+2.1 / +2.8	+1.9 / +2.7	+2.6 / +3.3
Tags, simple / trees	+0.9 / +0.4	+0.7 / −0.1	+1.5 / −0.3
5-number formula	+1.0	+0.6	−0.3
Tobler 1993	−12.5	−12.0	−9.2

Zones: Northeast, Southern Appalachians, Rockies, Pacific Northwest, California. Terrain beats tags in every bootstrap draw, in all 5 zones and both directions. The 5-number formula travels less well.

Backup: season, time of day, weather



New regions, skill (%)	simple	trees
Season + time of day	−0.3	−1.3
Weather (Daymet)	−0.0	−1.9
Terrain / terrain + weather	+2.1 / +1.9	+2.8 / +1.4

Hot days (>25°C) are 3% slower and snow on the ground 4%, but the hiker's own pace that day already holds it.